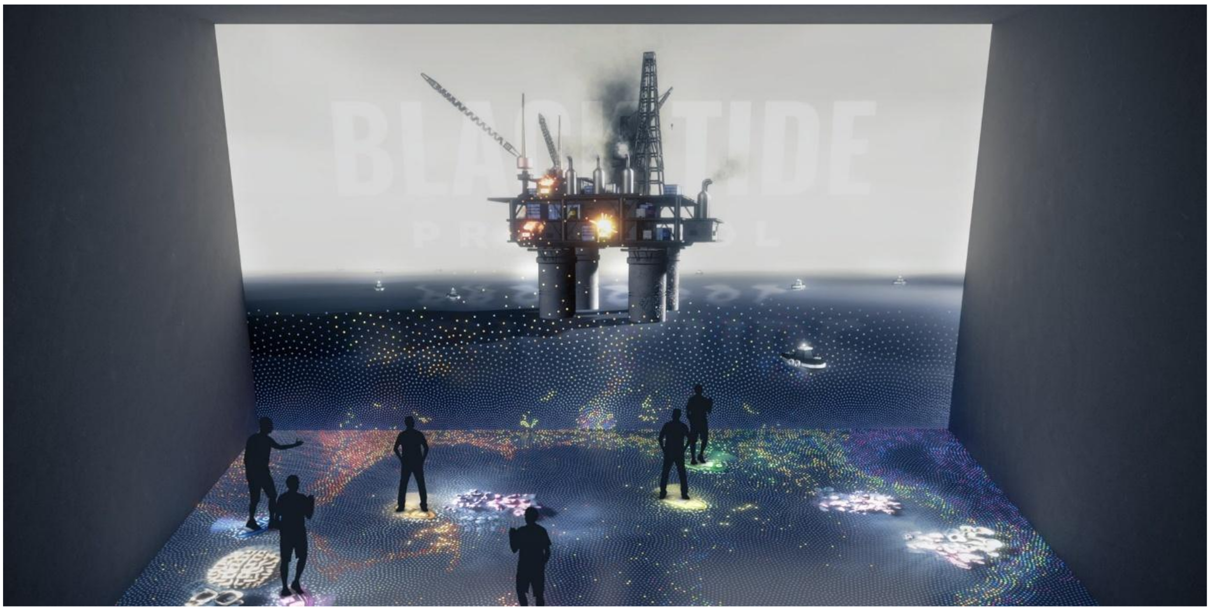

Black Tide Protocol

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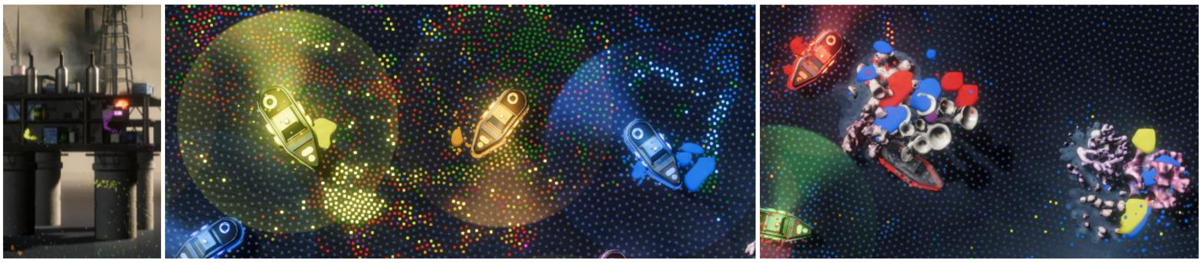
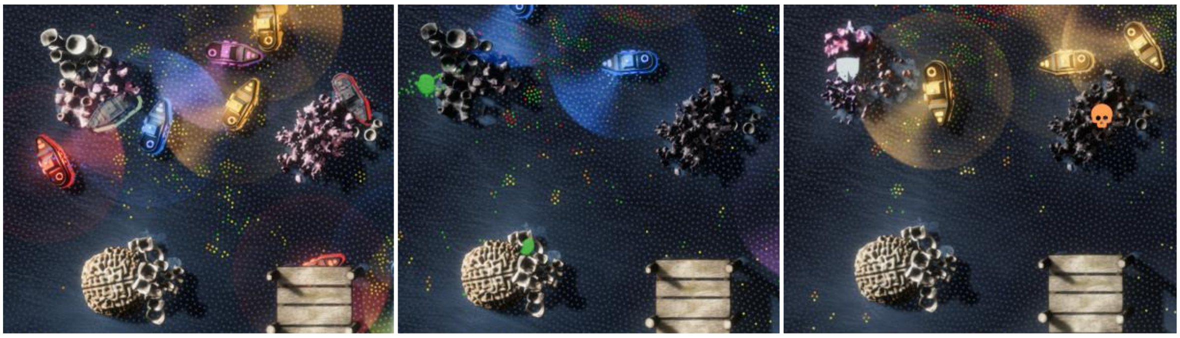
Black Tide Protocol was designed as a multiuser interactive experience for the Ars Electronica Center's Deep Space 8K and portrays an ecosystem on the brink of disaster. A defective oil platform spews a deadly, colorful tide into the ocean that threatens to wipe out an entire seal colony. Use the lights from your skimmer ships to track down absorbable oil particles and split complex oil mixtures before removing them. Can you stop the black tide before it is too late?

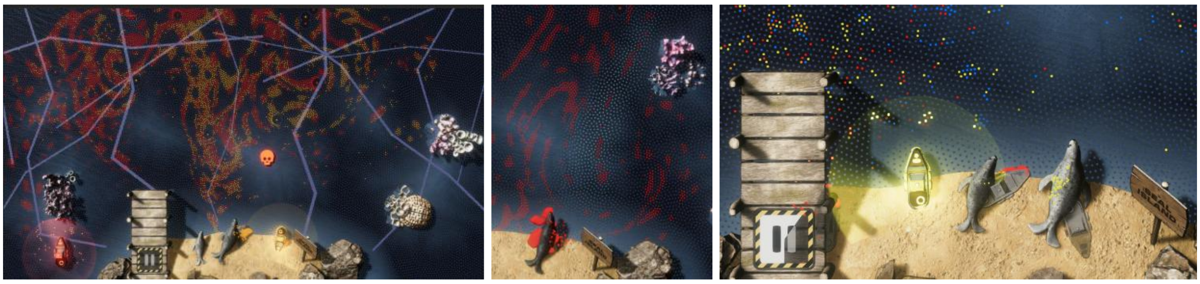
Base-Functionality:



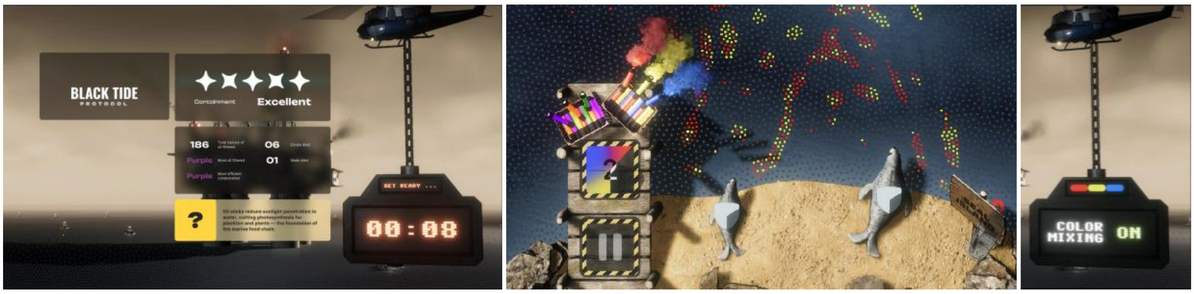
The game revolves around collecting matching oil particles, by moving over or near them, just as a boat would in real life. This has to be done in order to save the corals and seals placed on the interactive ground from getting killed by the oil (continuously spilled into the sea by the exploding oil barrels). To provide visual feedback of an oil removal, every oil particle removed from the simulation immediately creates a blob of the same color as the particle that hit the object. The light circle around the boat shows the highlighting distance (not the collection distance which is 2/3 smaller!), in which the collectible particles are brighter to help the players understand what they can collect next in real-time.



With each hit (regardless if the hit object is a coral or a seal), the color of the animal fades more towards black, indicating the object is one step closer to dying. If it has been hit too many times, it is removed and a small skull icon appears, indicating the animal's final death. After a few seconds, the animal respawns (with a temporary shield) at one of the 7 fixed spawn points and must be saved from being killed again.



To encourage interaction across the whole room, dynamic currents occur (also shown by a lighter coloring of the water particles) which force is based on the number of people near the wall) making it harder to protect the animals (and forcing the players to keep away from the wall for higher immersion). As an indicator that a player is out of bounds (colliding with/underneath the animals or island), the color of the boat (except for its' outline and if the boat is spawned in the exact frame) is set to grey. The simulation also can be paused when standing on the Pause Button (located on the pier of seal island) for more than 7 seconds, totally freezing the whole simulation (indicated by the boats frequently blinking). The simulation will immediately resume once the player steps off the button.



At the end of the round (usually after four minutes), the final score is calculated (the simulation will continue normally though, but not capturing any points anymore) and will be shown in detail to the audience before the game automatically restarts after a set amount of time (usually 40 seconds, shown on the timer hanging from the helicopter).

During the end screen, the Color Mode Button will appear on the pier of seal island. Stepping on this button for 5 seconds allows to toggle the Color Mode, quickly switching between Single Color Mode and Color Mixing Mode (shown by the colors of the field and the activated colored rockets - showing all available boat colors for the next mode). To switch the Color Mode once again, the player needs to step off the button before stepping onto it again – which is shown by the timer reappearing. After the mission is finished, the simulation will automatically restart in the last chosen Color Mode (which was shown on the display hanging from the helicopter).

Colors, Mixing and Boat Colors:

The entire game is based on the primary and secondary colors of the standard color wheel with the respective ordering: Red, Yellow, Blue, Orange (Red+Yellow), Violet (Red+Blue), Green (Blue+Yellow).

This means that when a player joins the game, their boat color is determined by the next available and underrepresented color (the color used the least as boat color – in this case Violett, then Green). If this color is not yet available as a barrel or particle, at least one barrel and their corresponding particles immediately change color, so every player can interact with the simulation at any time (see Color-Specifications for technical details).

Another important point is that a player's color only changes when Pharus detects a physical exit from the field (e.g. jumping, leaving the tracked area, etc.) to ensure consistency and minimize confusion during play.

Color Modes:

Since the entire simulation is dynamic - including changing the available colors in real time based on the number of players, boats, barrels and particles - a description of each mode can be found below:

- ### Single Color Mode

  At the beginning of the game, the Single Color Mode is activated, meaning that no color mixing is possible. This makes all six colors available as boat colors, and the available colors simply change based on the number of players in the playing field.

- ### Color Mixing Mode

  Once activated, the boats will only use the three primary colors (Red, Yellow and Blue) to allow the mixing of the secondary colors for the particles. If two players with different colors are close to each other, a colored rope between the boats will appear to indicate that the specific particle color can now be collected by both players (also shown by the mixed particle color being highlighted around the players now). Once the two players move too far apart, the rope will disconnect to indicate that the mixed color can't be collected anymore.

Technical Details and Design Choices:

If anything remains unclear, the whole source code is available in the Github-Repository which can be found here: https://github.com/luca-g97/BlackTideProtocol-DeepSpace.
The folder of interest (whole simulation code) is located at: ...\Assets\Scripts\Sim2D and divided into Wall and Floor, which are similarly structured.

Setup:

Two physically accurate fluid simulations are interconnected between the two screens, simulating the spillage of oil into the sea from the user's perspective. This also enables different settings to be used for each simulation, which is necessary to guarantee gravity-based falling behaviour on the wall and make the simulation as realistic as possible.

The area of the playground can dynamically be set by changing the play area in Pharus, ensuring that the simulation will immediately adapt to the new boundaries.

Color-Specifications:

The color displayed for both the barrels and the particles is completely different to the actual colors. This means that the color underneath the displayed color of the particles and barrels always stays the same. This enables the colors to change dynamically depending on the number of players, because if a color is not yet present as a boat, the next available color is shown instead. This enables any number of players, from one to a theoretical maximum of 1024, to play the game simultaneously.

Potential Issues:

When changing back from Color Mixing Mode to Single Color Mode, any boats already present will keep their colors. Only the three primary colors will be shown until a new player enters the scene and receives the next available color (usually orange). This is intentional, as the player ID usually changes when a new player joins, which would randomly change the colors of the other players. This prevents the colors from being reassigned as previously saved whenever a player enters or leaves the scene.


Another issue is that sometimes a horizontal current appears for a short amount of time in the middle of the ground projection. This is not intentionally, but can't be prevented, since the cause is located inside Pharus and not within the simulation (usually only if the Pharus data is broadcasted).

Whenever no player is detected by Pharus anymore, the whole simulation stops immediately and every color switches to white (except for the particles in the color of the player who left the last) until a player is detected again.

Individual Game-Settings:

Simulation-Settings:

These can easily be individually set in the Simulation-Config located at:
"...\Assets\StreamingAssets\SimulationSettings.xml"

Use the XML settings to override the standard settings in the game:

<useXML>true</useXML> → XML overrides the standard settings

Setting of the bounds (screen space) which is usually set to 16:9:

<boundsX>16</boundsX> → Ground-Screen X-dimension

<boundsX_Wall>16</boundsX_Wall> → Wall-Screen X-dimension

<boundsY>9</boundsY> → Ground-Screen Y-dimension

<boundsY_Wall>9</boundsY_Wall> → Wall-Screen X-dimension

Simulation-Settings - Change with caution, since this can totally break the simulation:

<timeScale>1</timeScale> → General time scale for the Ground-Screen Simulation

<timeScale_Wall>1</timeScale_Wall> → General time scale for the Wall-Screen Simulation

<fps>144</fps> → Simulation-FPS for the Ground-Screen Simulation

<fps_Wall>144</fps_Wall> → Simulation-FPS for the Wall-Screen Simulation

<iterationsFrame>3</iterationsFrame> → Simulation-Iterations per frame for the Ground-Screen

<iterationsFrame_Wall>3</iterationsFrame_Wall> → Simulation-Iterations per frame for the Wall-Screen

<gravity>0</gravity> → Gravity-Setting for the Ground-Screen Simulation

<gravity_Wall>-75</gravity_Wall> → Gravity-Setting for the Ground-Screen Simulation

<totalParticles>100000</totalParticles> → Maximum Particles for the Ground-Screen Simulation

<totalParticles_Wall>100000</totalParticles_Wall> → Maximum Particles for the Wall-Screen Simulation

Current-Settings – Change with caution, since this can heavily affect the simulation:

<sidewaysSpawnVelocity>200</sidewaysSpawnVelocity> → Spawn-Velocity of the particles for the Ground-Screen Simulation

<maxDynCurrentStrength>33</maxDynCurrentStrength> → Maximum-Strength of the dynamic currents

Difficulty Settings:

These can easily be individually set in the Difficulty-Config located at:
"...\Assets\StreamingAssets\DifficultySettings.xml"

Difficulty-Settings – Changing the health of the animals and the strength of the currents based on player amount:

<basePlayerCount>4</basePlayerCount> → Player-Amount to dynamically setup the settings

<maxDifficultyMultiplier>2</maxDifficultyMultiplier> → Maximum-Multiplier for the difficulty of the whole game *(2 times harder when a lot of players are present)*

<minDifficultyMultiplier>0.5</minDifficultyMultiplier> → Minimum-Multiplier for the difficulty of the whole game *(2 times easier when only one player is present)*

Mission-Settings:

These can easily be individually set in the Mission-Config located at:
"...\Assets\StreamingAssets\MissionSettings.xml"

<totalMissionRuntime>240</totalMissionRuntime> → Time in seconds until the mission ends and the End-Screen is displayed

<missionOvertime>10</missionOvertime> → Time in seconds until the final result is shown

<missionRestartDelayAfterGrade>40</missionRestartDelayAfterGrade> → Time in seconds to wait until restarting the mission

<coralPenalty>10</coralPenalty> → Penalty for letting a coral die

<sealPenalty>50</sealPenalty> → Penalty for letting a seal die

<mixedOilCountPerPointRecovered>100</mixedOilCountPerPointRecovered> → Number of mixed oil particles (colors: Orange, Green and Purple) need to be filtered to recover 1 penalty point